

# Changjin He

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*References available on request*

## Summary

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Mechatronics engineer with four years of client-facing field engineering experience in the semiconductor industry, complemented by a recently completed Master's degree and hands-on experience delivering data platforms and analytical research. Combines hypothesis-driven quantitative analysis, including statistical root-cause modelling and controlled experimental design in R and SQL, with a track record of solving complex technical problems on-site and communicating clearly with clients and stakeholders across two countries.

## Skills

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- **Analytical:** Statistical analysis (R: ANOVA, FDR correction), SQL (Vertica, DuckDB), Python, experimental design, hypothesis testing, data processing and visualisation
- **Technical Delivery:** FastAPI, React/TypeScript, Docker, Azure, Git, CI/CD workflows
- **Communication & Collaboration:** Stakeholder handoff, customer-facing technical support, teaching, technical documentation

## Professional Experience

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### Auckland Transport

*Computer Vision Intern (Full-time)*

**Nov 2025 – Feb 2026**

*Auckland, New Zealand*

- Designed and developed an internal operations dashboard enabling transport teams to query, process, and analyse operational data, improving workflow visibility and supporting day-to-day operational decisions.
- Tested AI vision models against real-world transport data and documented validation findings, helping teams understand where model outputs could and could not be trusted for decision-making.
- Worked in an Agile delivery environment, documenting technical issues and deployment behaviour to support reliable handoff between software, data, and operational stakeholders.

### Cloudcell

*Software Development Intern*

**May 2025 – Oct 2025**

*Auckland, New Zealand*

- Conducted market research on affordable drug alternatives and translated findings into the design of a data platform that helped users search, compare, and analyse large-scale chemical compound data.
- Developed a drug similarity search platform using FastAPI, DuckDB, RDKit, and React/TypeScript. Processed approximately 2.47 million compound records from ChEMBL 35, built a local drug structure database, and pre-computed Morgan fingerprints and molecular physicochemical properties. Supported SMILES/drug name search, Tanimoto/Cosine similarity search, molecular property filtering, structure visualization, and result export.
- Integrated the seyonec/ChemBERTa-zinc-base-v1 model to generate 768-dimensional molecular embeddings, enabling AI-powered molecular similarity search based on semantic representations. Designed a candidate drug post-processing workflow, including composite ranking, clustering, and drug pricing data aggregation, to support the discovery of potential low-cost drug alternatives.
- Delivered user-facing functionality iteratively through structured testing and documentation.
- Demo: [ccccjin.github.io/Affordable-Drug-Alternatives](https://ccccjin.github.io/Affordable-Drug-Alternatives)

### University of Auckland

*Graduate Teaching Assistant (Part-time)*

**Mar 2025 – Jun 2025**

*Auckland, New Zealand*

- Supported MECHENG 313 labs and tutorials involving explaining embedded systems, troubleshooting steps, and practical engineering procedures clearly to students with different technical backgrounds.

### Axcelis Technologies, Inc.

*Field Service Engineer*

**Jun 2023 – Jun 2024**

*Shenzhen, China*

- Provided on-site installation, diagnosis, troubleshooting, maintenance, and repair support for complex semiconductor equipment at customer sites, testing and certifying equipment operating performance and system quality.
- Delivered customer training on system operation and maintenance, translating complex technical procedures into clear, usable guidance for client teams.
- Owned end-to-end field service activities spanning hardware, software, PC, network, and wireless network systems support.

### Nikon Precision Shanghai Co., Ltd

*Field Service Engineer*

**Jul 2019 – Aug 2022**

*Shenzhen, China*

- Diagnosed complex equipment failures through root-cause analysis of system logs and waveform data, applying Fourier Transform analysis and designing feedforward, notch, and peak filters to optimise system performance.
- Identified causes of precision overlay anomalies using a  $3\sigma$  statistical model of linear and nonlinear compensation values, comparing interferometer brightness and temperature data across time periods to isolate contributing factors.
- Calibrated laser heads and optical paths using a collimator (Zygo DynaFiz), and collected waveform logs via a multi-channel data acquisition system (Keyence NR-500) to perform logical analysis of vibration causes.

## Research & Projects

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### SO-ARM101

*Independent R&D Engineer*

**Jun 2026 – Now**

*Auckland, New Zealand*

- Developed hands-on experience in robot teleoperation, imitation learning, reinforcement learning, ACT, and Vision-Language-Action (VLA) models using the SO-ARM101 robotic platform, covering the full development pipeline from 3D printing and hardware assembly to software deployment.
- Completed a self-directed “Train LLM from scratch” project to understand large-model training mechanics, demonstrating rapid self-driven learning in unfamiliar domains.

### Emotion-Aware Assistive Robotic Arm

*120 Points Research*

**Jan 2025 – Feb 2026**

*Auckland, New Zealand*

- Developed a ROS/MoveIt-based UR5e pick-and-place control system with joint-space and Cartesian trajectory execution, FK computation, path planning, and simulation validation.
- Integrated UR5e, Robotiq 85 gripper, and conveyor belt ROS interfaces for real-robot control, controller switching, parameter tuning, and Sim-to-Real deployment.
- Built an emotion-aware human-robot collaboration system that used facial expression recognition to adapt UR5e speed and interaction distance.
- Designed controlled experiments on robot speed and distance, and evaluated user experience using GodSpeed, NASA-TLX, and task performance metrics.
- Built an R-based analysis pipeline to clean experiment data, compute performance/workload metrics, and run repeated-measures ANOVA, paired t-tests, EMM, and FDR-corrected correlation analysis.
- Automated statistical tables and visualizations to identify how distance and task type affected task time and robot perception.

### Emotion-Driven Interactive Education Chatbot

*Frontend development (university project)*

**Jul 2024 – Nov 2024**

*Auckland, New Zealand*

- Built a real-time emotion detection chatbot using OpenCV for video capture, YOLOv8 for face detection, and AlexNet (PyTorch) for emotion recognition.
- Implemented GUI for live video display with emotion labels at 20ms refresh intervals.

### Two-Arm Rubik's Cube Robot (C++)

*Intelligent Robot Creative Competition*

**2017 – 2018**

*Hangzhou, China*

- Built C++ motion-control logic for a two-arm Rubik's Cube robot, translating solver outputs into stable manipulator action sequences.
- Developed sinusoidal trajectory planning for controlled robot motion and reliable physical manipulation.
- Implemented camera-based colour recognition using Gaussian filtering and RGB-to-HSV conversion for physical cube-state classification.
- Strengthened early experience in C++ control logic, automated decision workflows, and hardware-software integration.

## Education

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### University of Auckland

*Master of Mechatronics Engineering (Graduated)*

**Jul 2024 – Jun 2026**

*Auckland, New Zealand*

- Main Courses: Human-Robot Interaction (A), Robotics and Society (A), Multivariable Control System, Advanced Control.
- Thesis: An Experimental Investigation of Robot Speed, Proximity, and Task Pacing in Industrial Human-Robot Collaborative Assembly (A-)

### Beijing Institute of Technology

*Bachelor of Mechatronics Engineering*

**Sep 2015 – May 2019**

*China*

- Main Courses: Mechanical Design and Theory, Programming C, Digital and Analog Electricity, Control Engineering, Theory and Applications of Embedded Systems, Machine Vision Technology. GPA: 2.82 (Top 30%)